THE SI SENTINEL

Weekday briefing on builders shaping super intelligence

THURSDAY, OCTOBER 22, 2026

THE MORNING PICTURE

Patch day for the web; Tesla and Meta report

Two critical flaws in PHP, the language behind roughly 70% of websites, went public with patches on Wednesday after a two-month effort that OpenAI says it led and that AI models helped start. Greg Brockman announced the coalition, Guillermo Rauch described Vercel's firewall rules going live before disclosure, Aaron Levie called cybersecurity the clearest example of the jobs AI will create, and Vitalik Buterin warned that finding bugs first is not a game defenders win by default.

Tesla reported third-quarter revenue of $28.05 billion and started Optimus production in Fremont; Ashok Elluswamy launched a Robotaxi safety page. Meta's results put its Muse agent at the center, and Alexandr Wang celebrated. Elon Musk said SpaceX has a shot at more than 10 gigawatts of compute next year. Twenty-four of the 46 builders posted; Musk posted 50 times.

I · PATCH THE PLANET

php2shell: AI found it, the industry raced to fix it

Two memory bugs in the PHP engine, present for a decade, allowed code to run on servers without a login. Patches shipped Wednesday morning, a day after details leaked.

9.2	5.4+	$1B
Severity score PHP gave both flaws (CVSS 4.0)	Every PHP version since 5.4 affected by at least one bug	Subsidized defender access OpenAI pledged alongside the fix

Greg Brockman @gdb · President, OpenAI

Brockman wrote at 10:58am EDT that "over the past two months, we have been leading a coalition to mitigate one of the most severe bugs in open source software found in recent memory, php2shell, found as part of our Patch the Planet initiative," linking an X article. OpenAI's write-up says the security firm Calif found a critical flaw using GPT-5.6 Sol, Wiz independently found and reported both, and PHP's maintainers led the fix. OpenAI says it worked with the White House's Super Intelligence Force, the Office of the National Cyber Director and the Cybersecurity and Infrastructure Security Agency (CISA), committed $1 billion in subsidized access to its Daybreak security tools for under-resourced defenders, and proposed a five-stage framework for disclosing AI-found bugs. "These bugs are among several significant vulnerabilities we've already found," it wrote, "and we expect similar findings to increase sharply in the coming months."

The flaws, CVE-2026-109618 and CVE-2026-109619, sit in how PHP cleans up memory after an out-of-memory or fatal error; a crafted request can make a later request reuse stale or half-built data. Wiz Research said it chained them into one script that ran code without authentication on stock WordPress, Drupal and Joomla images, and that frontier models "substantially reduced" a barrier that sat "at the edge of what human researchers could practically achieve." Wiz found the first bug on August 19 using Gemini 4 Argon and GPT-5.6 Sol. Details leaked on October 20, and PHP shipped 8.5.12, 8.4.27, 8.3.36 and 8.2.35 around 9:30am EDT Wednesday, according to Huntress. TechTarget counted 22 million exposed servers. The National Security Agency's cyber account advised that if an immediate update is not feasible, organizations should "take the system offline until it can be fully mitigated."

Guillermo Rauch @rauchg · CEO, Vercel

Vercel's bulletin said its web application firewall (WAF) rules shipped before disclosure. Cloudflare said its own rules went live about nine hours ahead. Rauch, at 3:29pm EDT:

> *"It was inspiring to watch the behind-the-scenes of the industry coming together, weeks in advance, to protect the world's infrastructure and mitigate php2shell. We deployed Vercel WAF mitigations that have already successfully prevented attempts on PHP origins. Assume more vulnerabilities are on the way. Flex the patch → redeploy muscle."*
>
> GUILLERMO RAUCH · @RAUCHG · 3:29PM EDT · POST

More arrived within hours. A high-severity remote-code flaw in Node.js's Brotli decoder hit versions 22, 24 and 26, and at 7:29pm EDT Rauch wrote that "Vercel has proactively rolled out Firewall-level mitigations for today's Node.js RCE vulnerability, in advance of the official patch release. All traffic for all deployments is automatically protected," pointing to Vercel's advisory.

Tobi Lütke and Aaron Levie @tobi · CEO, Shopify · @levie · CEO, Box

Tony Ma posted a tally of one day's disclosures: two PHP flaws, ten in Node.js, five in NGINX, one in Go, twelve in GitLab and a certificate-authority compromise at eMudhra. Lütke quoted it at 11:17pm EDT: "Craziest day in cyber security in a long time." Levie had quoted the NSA advisory at 1:48pm EDT:

> *"Cyber is going to be one of the most profound aspects of AI. We now have a tireless workforce that can hunt down, discover, patch, or attack any system and software ever created. ... We're entering a new cyber regime, and we'll probably use 100X more tokens on this loop than we're imagining right now, because we can scan every log, look at every line of code, look through traffic, and process data at a scale that wasn't comprehensible before."*
>
> AARON LEVIE · @LEVIE · 1:48PM EDT · POST

At 1:15am EDT he added: "If you wanted to know all the jobs that AI will produce, look no further than cybersecurity. ... even as the work becomes more agentic and autonomous you will need more talent to point the agents at the most critical work, review findings, patch systems, respond to incidents, and more."

Vitalik Buterin @VitalikButerin · Co-founder, Ethereum

Nate Soares, president of the Machine Intelligence Research Institute, called it "striking" that the flaw was found by Sol and Argon rather than fixed after the Mythos models six months ago, "a heavy strike against the 'defense will run ahead of offense' theory." Buterin answered at 9:22pm EDT:

> *"I didn't predict and do not predict defense running ahead of offense automatically. I predict cybersecurity will be endgame defense-favoring but the transition will be hard. In particular, "good guys find vulns in already-released software before the bad guys do" is not a defense-favoring game. Making the game defense-favoring requires deeper shifts in methodology including formal verification, aggressive TCB minimization, etc, and we've only barely started making those transitions."*
>
> VITALIK BUTERIN · @VITALIKBUTERIN · 9:22PM EDT · POST

TCB is the trusted computing base, the code a system must trust to be secure. Buterin added that this is "a big reason why I continue to be friendly toward slowdowns/pauses, and we need >1% of GDP going toward cyber + bio + physical world hardening." OpenAI's post makes a similar case for memory-safe rewrites, formal verification and tighter sandboxes.

QUICK HIT

Paul Graham (@paulg), Y Combinator co-founder, to a reminder that October is cybersecurity awareness month: "The future is cybersecurity awareness future."

II · TESLA AND SPACEX

Tesla's quarter, a Robotaxi safety page and 10 gigawatts

Tesla reported after Wednesday's close and began building Optimus in Fremont. Musk spent the evening on FSD demand and SpaceX's compute plans.

$28.05B	1.74M	3.3M
Tesla Q3 revenue, flat with a year earlier	Active FSD subscriptions, up 67% year over year	Autonomous Robotaxi miles, with zero injuries reported

The numbers Tesla (TSLA) · quarter ended Sept. 30

Tesla's third-quarter update showed revenue of $28.05 billion against $28.10 billion a year earlier, GAAP net income of $787 million ($0.22 a share) and non-GAAP net income of $1.70 billion. Operating cash flow was $6.3 billion and free cash flow $1.5 billion. Tesla said Full Self-Driving (FSD) (Supervised) was attached to more than 75% of new North American deliveries, a gain of nearly 20 points in a quarter, and that battery supply, not demand, limited vehicle volume; it ended with its largest passenger-car backlog in four years. Cybercabs began carrying Robotaxi riders in Austin on September 3, and unsupervised miles grew eightfold to nearly 2 million. Optimus entered

production on high-volume lines in Fremont, which Tesla calls "the first production-quality bots we've built"; on the call, Musk said reaching 20,000 robots a week would probably take at least a year, a transcript summary shows.



A gold Tesla Cybercab on Interstate 280 near Palo Alto, California, on September 19, 2026.

PHOTO: DANIEL LU (DLLU), "TESLA CYBERCAB ON INTERSTATE 280 NEAR PALO ALTO," WIKIMEDIA COMMONS · CC BY-SA 4.0 · CROPPED

Quoting Sawyer Merritt's summary, which put active FSD subscriptions at 1.74 million against 1.04 million a year earlier, Musk wrote at 4:54pm EDT, "People are buying FSD with a car attached, not the other way around." On Tesla's own highlights post he wrote, "Great progress by the Tesla team!" After the commentator Farzad graded SpaceX's earnings production "A+" and Tesla's "F-," Musk replied at 11:36pm EDT, "Fair enough, we will make it more entertaining next time."

Ashok Elluswamy @aelluswamy · Head of AI software, Tesla

Hours before the report, Elluswamy introduced a public Robotaxi safety page at 2:58pm EDT:

> *"We are launching the Tesla Robotaxi safety report today. Till date, Robotaxi has driven a total of 3.3 million miles autonomously, and has had zero injuries nor airbag deployments. If the same number of miles were to be manually driven, we should expect roughly 9 injuries to the occupants or other pedestrians and 4 airbag deployments on average. We will continue to transparently share our progress with the public using this page."*
>
> ASHOK ELLUSWAMY · @AELLUSWAMY · 2:58PM EDT · POST

Elon Musk @elonmusk · CEO, SpaceX and Tesla

A day after SpaceX's own results, a clip circulated of the investor Brad Gerstner saying Musk had described standing up 200,000 GB300 chips a month, 3 gigawatts of compute by year end and 10 gigawatts next year, about a quarter of the 43 gigawatts forecast for the whole United States. Musk replied at 9:42pm EDT:

> *"We have a shot at exceeding 10GW next year"*
>
> ELON MUSK · @ELONMUSK · 9:42PM EDT · POST

To the author Tae Kim, who read Musk's call remark that hyperscalers relentlessly ask to rent SpaceX compute as bullish for smaller cloud providers, Musk answered at 10:15am EDT: "Only if they can get their compute online and reliable, which is extremely difficult to do at scale." He called a Morgan Stanley note describing SpaceX's cloud build as a "funding strategy" for training its own robots "Interesting."

> QUICK HIT
>
> **Starlink on United.** United's chief executive said customers are switching from other airlines for Starlink Wi-Fi, which will reach half its flights this week. Musk, at 12:54pm EDT: "Starlink moves the needle."

Uncle Bob Martin (@unclebobmartin), the software author, explained FSD's driver monitoring to a skeptic: it "allows you to take your hands and feet off the controls, but does not allow you to ignore the road. The internal camera makes sure your eyes are pointed out the front," he wrote.

III · THE AGENT RACE

Meta bets the company on Muse; Google and Microsoft answer

Personal agents dominated earnings day. Meta said Muse reached 10 million users in six weeks, Sundar Pichai described Google's plan, and Satya Nadella published his annual letter.

Alexandr Wang @alexandr_wang · Chief AI officer, Meta

Meta reported third-quarter revenue of $64.6 billion, up 26%, with free cash flow of negative $2.12 billion after $34.91 billion in capital spending, and announced a non-voting Class C stock to fund its AI build-out. On the call, Mark Zuckerberg said more than 10 million people in the US and Canada set up Muse agents in a month and a half, about 25% use it on a given day, and shopping is the top use, with average purchases over $200. When Kalshi posted that Zuckerberg said Muse could earn more revenue per user than Meta's family of apps, Wang quoted it at 5:08pm EDT with a link. To a post that Zuckerberg had said "muse" about 50 times in ten minutes, he wrote at 7:44pm EDT: "this was the gameplan."

Alyson Shontell wrote that after two weeks with Muse and a day with a rival agent, Instinct, she liked Muse better, partly because it "seems better at proactively bringing me info that's important that I might have missed." Wang replied at 7:56pm EDT: "ya i mean i really trust ur vibe and if ur saying u like muse better then who am i to argue."

Sundar Pichai @sundarpichai · CEO, Google and Alphabet

Pichai promoted a long interview with Alex Heath at 12:34pm EDT: "Great to sit down with @alexeheath to talk about Argon, the frontier, and our plans for personal agents!" Heath's notes say it covers the reception of Gemini 4 Argon, the recent leadership changes at Google DeepMind, Google's free cash flow turning negative, space data centers, and whether agents will tie Search, Gmail and Gemini together. Heath's summary: "Even Google can't get enough compute."

Satya Nadella @satyanadella · CEO, Microsoft

At 4:45pm EDT Nadella posted his annual letter, on "building a frontier ecosystem to turn technological progress into real-world outcomes and economic opportunity." Overnight he highlighted customers: Volkswagen Group has given Microsoft Copilot to more than 330,000 employees ("Super to see Volkswagen Group putting Copilot to work across the company!"), and in Cologne he heard how TK Elevator gives service technicians agent access to decades of shared repair knowledge, Microsoft's story says.

> QUICK HIT
> **Elon Musk** pitched Grok Bot all day: show it last year's tax returns because "There may be refunds you're owed," and "Grok @Bot is like hiring a super smart and fun employee!"

Patrick Collison (@patrickc), Stripe's chief executive, at 1:44pm EDT: "Ben has for many years been the best analyst in tech. I'm very much looking forward to this new @stratechery + @stripe partnership." Ben Thompson's announcement says Stratechery is now "presented by Stripe," with total editorial independence and a disclosure in every relevant post.

IV · MODELS AND BUILDERS

Fast open models, cleaner agent code and a $1 billion bet on GPUs

DeepSeek's latest small model set speed records on hosted chips. Builders showed what agents now ship, and how to keep their code clean.

David Heinemeier Hansson @dhh · Creator of Ruby on Rails and Omarchy

DeepSeek released V4.1-Flash-1017 on October 17 as open weights under the MIT license, with a one-million-token context. Fireworks said it runs at more than 1,521 tokens per second on its fastest tier, "7x faster than Claude Opus 5.5." Heinemeier Hansson quoted that at 10:12pm EDT:

> *"I remember when 200tps felt impossibly fast. This is going to be crazy."*
> DAVID HEINEMEIER HANSSON · @DHH · 10:12PM EDT · POST

Rauch added the model to Vercel's Token Pass at 4:50pm EDT, calling it "a phenomenal model to run subagents that execute plans conceived by larger and more expensive models." Heinemeier Hansson also cheered Omarchy, his Linux setup, booting on an M6 Mac Mini after five days of work ("We can fix everything. Even the very latest M chips!") and running natively on a sixth-generation iPad ("Wow!").

Uncle Bob Martin @unclebobmartin · Software author

Martin described his workflow at 10:59am EDT: "I talk the system over with my agents, they develop a spec. They build the system (that would have taken me a week to build) in an hour or two. Then I have them run the crapper, dryer, and mutator checks." Asked whether he still reads the code, he said at 11:04am EDT:

> *"Humans are slow at code. Agents are fast. So reading code is a liability. However, we still need to ensure that the code is clean enough for the agents to continue to manage and maintain. This can be done by using a set of deterministic tools that measure the quality of the code and how well tested the code is."*
> UNCLE BOB MARTIN · @UNCLEBOBMARTIN · 11:04AM EDT · POST

The tools are on his GitHub: crapper scores complexity against test coverage, dryer finds duplication and mutator runs mutation tests. Asked why agents write bugs at all, he replied at 6:06pm EDT, "Coding is hard, even for SI. But double and triple checking works pretty well. Unit tests work even better. Mutation tests work even better than that."

Jeff Dean and Michael Truell @JeffDean · Chief scientist, Google · @mntruell · Co-founder, Cursor

Flapping Airplanes, a year-old lab that rebuilt the GPU software stack around "megakernels" running whole programs as one kernel, raised a $1 billion Series A co-led by Index Ventures and Lux, Index wrote, at a $6 billion valuation, TechCrunch reported. Dean wrote "Congrats to the whole team"; Truell, "Congrats Ben!!" to chief executive Ben Spector.

QUICK HIT

Greg Brockman pointed to GPT-Rosalind's lead on Perry syndrome, from thousands of agents reading mouse brain-cell data in 32 minutes: "Rosalind for new drug discovery ideas."

Tim Sweeney @TimSweeneyEpic · CEO, Epic Games

Sweeney opened Unreal Fest in Shanghai and won in court. Australia's Federal Court ordered Apple to readmit Fortnite and, by March 1, 2027, open iOS there to rival app stores, direct downloads and alternative payments, AppleInsider reported; Apple told AAP it will seek a stay and appeal.

Sweeney, 8:06pm EDT: "The Australian Epic v Apple antitrust remedies were just issued, and are a huge victory for consumers and developers." Of Benj Edwards porting Sweeney's 1991 game ZZT to the Nintendo Entertainment System with Claude Code, he wrote "Truly a new age of gaming and software ahead."

Clément Delangue (@ClementDelangue), Hugging Face's chief executive, noted Jensen Huang closed Nvidia's European tour with Microduck robots, designed in Bordeaux by Pollen Robotics and Hugging Face, "the most popular robot of 2026 with over 15,000 sold."

Also on the wire

Palmer Luckey. The Anduril founder answered a critic of his views on China at 4:26pm EDT: "I have manufactured billions of dollars of hardware in China and employed hundreds of Chinese engineers." Defending his case for underground military systems, he wrote that "autonomy+narrow diameter payloads make the concept feasible vs older manned subterrene concepts."

Recursive SI's Flint. Richard Socher's lab released Flint, an open-weight local coding model it says runs at more than 125 tokens per second on a MacBook Pro and was built in under four months by its automated research system. Clément Delangue cheered the launch on Hugging Face.

Guillermo Rauch on Pi. At 11:49am EDT he backed the open agent runtime Pi: "Giving developers and companies choice of model, control over the runtime, data, skills, plugins." Later he urged teachers to have a coding agent build explainers: "Grab a coding agent, ask it for an explainer, deploy it, share the link."

Uncle Bob Martin. Asked for his audit setup, he posted the full text of "ub-mode," an agent skill that orders every change as "First Make it Work. Then Make it Right. Follow Practice," with test-first development, mutation testing and a running log of every action.

Satya Nadella. Harry Stebbings marked 12 years and 3,000 episodes of his 20VC podcast with a Nadella interview; Nadella replied at 3:16pm EDT, "Was great to finally join you on the pod, Harry!"

Tim Sweeney. Asked when Fortnite returns to macOS in Australia: "Upon global victory." To a user who called the ruling better than the European Union's app law, he wrote it was "Definitely a better win than actual EU enforcement of EU law, which has dragged on for years."

David Heinemeier Hansson. Overnight he said Omarchy 4.5 already has "Snapdragon support for several machines" and that there is "an open PR" adding DeepSeek's agent harness.

Lisa Su. AMD's chief executive hosted Mercedes driver Kimi Antonelli before the Austin Grand Prix: "We'll be cheering for you at the track this weekend!"

Takeaways

1. AI models found php2shell, a decade-old pair of PHP flaws; OpenAI led a two-month coalition and says more such findings will "increase sharply."
2. Defenders moved first this time, with firewall rules live before disclosure, but Vitalik Buterin argues lasting defense needs formal verification and smaller trusted code.
3. Tesla's FSD attach rate topped 75% in North America and Optimus entered production; Robotaxi reports zero injuries over 3.3 million miles.
4. Meta says Muse reached 10 million users in six weeks; Google and Microsoft are pitching agents of their own.
5. Fast open models such as DeepSeek V4.1 Flash are becoming the workers beneath larger planning models.

On the beat

POSTED · 24

Elon Musk, Uncle Bob Martin, David Heinemeier Hansson, Guillermo Rauch, Tim Sweeney, Alexandr Wang, Satya Nadella, Palmer Luckey, Clément Delangue, Joe Gebbia, Tobi Lütke, Ashok Elluswamy, François Chollet, Paul Graham, Greg Brockman, Aaron Levie, Vitalik Buterin, Patrick Collison, Michael Truell, Jeff Dean, Tom Brown, Lisa Su, Michael Dell, Sundar Pichai.

SILENT · 22

Marc Andreessen, Jensen Huang, Andrej Karpathy, Sam Altman, Demis Hassabis, Dario Amodei, John Carmack, Yann LeCun, Fei-Fei Li, Mustafa Suleyman, Andrew Ng, Jim Fan, Mira Murati, Mark Zuckerberg, Noam Brown, David Singleton, Ilya Sutskever, Kanjun Qiu, Nat Friedman, Arthur Mensch, Guillaume Lample, Mark Chen.

The watch list stands at 46.

The SI Sentinel is compiled from public posts on X and published reporting. Quotes are reproduced verbatim unless marked as translated or paraphrased; emoji in quoted posts are omitted.